

BMS Integration Specification

Smart Building Platform — Revision 2.1, 15 March 2026

1. Scope

This document specifies the integration of the **Building Management System (BMS)** with the cloud platform. The BMS monitors and controls HVAC equipment, lighting and power meters in the building.

2. System Components

- Air Handling Unit (AHU) controllers, model **FX-PCG2611-0**
- Variable Air Volume (VAV) terminal controllers
- Occupancy sensors and CO2 sensors (0–2,000 ppm)
- Edge device and gateway (BACnet/IP, Modbus TCP, MQTT)

3. Network Requirements

Item	Value
BMS server	192.168.1.100
BACnet/IP port	47808
MQTT broker	mqtt.example.com:8883
Polling interval	15 minutes

4. Alarm Handling

When a chiller alarm occurs, the system notifies the maintenance team by e-mail (support@example.com) and creates a work order. If the alarm is not acknowledged within 30 minutes, it is escalated to the facility manager.

Note: Setpoint changes require operator approval.

5. Energy Management

The Energy Management System (EMS) records energy consumption every 15 minutes. Peak demand was reduced by 18% in summer 2025 through demand response and a cooling setpoint adjustment from 24 °C to 26 °C.

6. Data Retention

Trend log data is stored for 13 months in the cloud platform at <https://bms.example.com/trends>.

